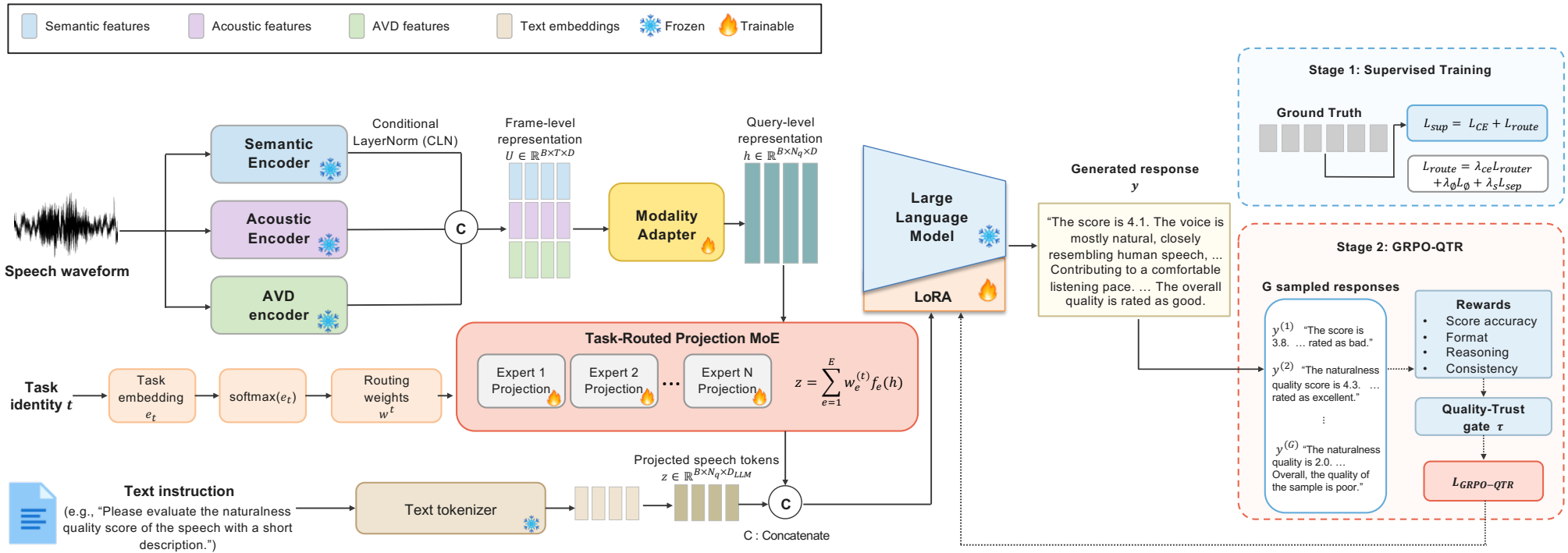

Semantic features
Acoustic features
AVD features
Text embeddings
Frozen
Trainable
Speech waveform
Semantic Encoder
Acoustic Encoder
AVD encoder
Conditional LayerNorm (CLN)
C
Frame-level representation
$U \in \mathbb{R}^{B\times T\times D}$
Modality Adapter
Query-level representation
$h \in \mathbb{R}^{B\times N_q\times D}$
Large Language Model
LoRA
Generated response
y
"The score is 4.1. The voice is mostly natural, closely resembling human speech, ... Contributing to a comfortable listening pace. ... The overall quality is rated as good.
Task identity t
Task embedding e_t
softmax(e_t)
Routing weights w^t
Task-Routed Projection MoE
Expert 1 Projection
Expert 2 Projection
Expert N Projection
$z = \sum_{e=1}^{E} w_e^{(t)} f_e(h)$
Projected speech tokens
$z \in \mathbb{R}^{B\times N_q\times D_{LLM}}$
Text instruction
(e.g., "Please evaluate the naturalness quality score of the speech with a short description.")
Text tokenizer
C : Concatenate
Stage 1: Supervised Training
Ground Truth
$L_{sup} = L_{CE} + L_{route}$
$L_{route} = \lambda_{ce} L_{router} + \lambda_{\emptyset} L_{\emptyset} + \lambda_s L_{sep}$
Stage 2: GRPO-QTR
G sampled responses
$y^{(1)}$ "The score is 3.8. ... rated as bad."
$y^{(2)}$ "The naturalness quality score is 4.3. ... rated as excellent."
$y^{(G)}$ "The naturalness quality is 2.0. ... Overall, the quality of the sample is poor."
Rewards
Score accuracy
Format
Reasoning
Consistency
Quality-Trust gate τ
$L_{GRPO-QTR}$